Mihael Cudic

Email: mcudic@mgh.harvard.edu

Education:

University of Oxford (UO), Balliol College, UK 2018-2023

Ph.D., Computer Vision for BioMedical Sciences

- Fully funded through the **National Institutes of Health Oxford-Cambridge Scholars Program** (1 of 11 in USA)
- Recognition: **National Science Foundation Graduate Research Fellow Recipient** (1 of 2,000 in USA)

University of Florida (UF), USA 2014-2018

B.Sc., Electrical Engineering

- Graduated through the Honors Program and Summa Cum Laude (GPA: 3.97/4.0)
- Fully funded through the **Lombardi Scholars Program** (1 of 11 @ UF)
- Recognition: **Barry Goldwater Scholar** (1 of 240 in USA), UF's **Outstanding Scholar** (1 of 32 @ UF), UF's **Electric E Award** (1 of 8 @ UF's ECE Dept.), **x8** UF's **Dean's List**

Relevant Skills:

Machine Learning: Neuroimage Analysis • Medical Image Analysis • Deep Learning • Computer Vision • Generative Adversarial Networks • Segmentation • Auto-encoders

Biostatistics: Genomic Analysis • Genome Wide Association Studies • Causal Inference • Target Trial Emulation • Stochastic Compartmental Models • Conformational Analysis

Wetlab: Fluorescent Labelling • Microscopy • Dissections

Research/Work Experience:

Massachusetts General Hospital (MGH), Boston, MA USA Sep.23-Present

Precision Psychiatry Postdoctoral Research Fellow & Center for Genomic Medicine T32 Fellow

Mentor: Dr. Jordan Smoller (Director of the Center for Precision Psych. at MGH)

- Developing statistical machine learning methods to calibrate polygenic scores to an individual's environmental context
- Performed a comprehensive analysis of polygenic score prediction accuracy across intersecting environmental contexts [2]
- Led the implementation of a target trial emulation examining depression outcomes from COVID containment policies for an international research consortium [1]
  (Additional Project Supervisors: Dr. S. Bauermeister, Dr. M. Hernan, Dr. V. Patel, Dr. R. Kessler)

University of Oxford, Oxford, UK **& National Institutes of Health**, Bethesda, MD USA Aug.18-Jun.23

Computer Vision and Neuroscience Graduate Research Fellow

Mentors: Professor Alison Noble OBE (Former Director of the Inst. Of BioMed. Eng. at OU) and Dr. Jeffrey Diamond (Current Scientific Director at Nat. Inst. Of Neuro. Diseases and Stroke)

- Created the first deep learning pipeline for fully automated Fluorescent Microscopy (FM) image analysis of neurons using Generative Adversarial Networks while also increasing segmentation accuracy by 25% [4]
- Increased analysis accuracy on low Signal-to-Noise Ratio FM images of neurons by 15% using a novel deep learning architecture that fuses local and global scales [3]

University of Florida, Gainesville, FL USA Jan.15-May.18

Undergraduate Researcher & University Scholar

- Improved small-sample image classification by up to 22% using a kernel-based machine learning algorithm [5]
- Developed a neuro-inspired computer vision model using gamma saliency maps to improve answering image-related questions by 6% [7, 8]

Carnegie Mellon University, Pittsburgh, PA USA Jun.-Aug.17

Machine Learning Intern & Summer Scholar

- Predicted genotypes of plants from in-situ images with 78% accuracy, demonstrating the potential of image-based analysis as a low-cost, real-time alternative to DNA sequencing [6]

Appointments:

- **Adjunct Honors Dunlevie Professor** at UF (**4.8/5.0** evaluation for Signature Course) Jan.-May24
- Undergraduate **Engineering Computation Teaching Assistant** at UO Oct.-Ex.Dec.22
- Graduate **Machine Learning Teaching Assistant** at UF Aug.-Dec.17
- Undergraduate **Electronic Circuits Teaching Assistant** at UF Jan.-May.16

Extracurriculars & Other:

Study Abroad:

- **University of Pretoria**, South Africa: *South African Politics & History* (GPA: 4.0/4.0) May-Jun.17
- **Universidad Autónoma de Yucatán**, Mexico: *Tropical Ecology & Sustainability* (GPA: 4.0/4.0) Jun.-Aug.14

Other Roles: UK's **Young Entrepreneurs Society Member** (2022); Balliol Graduate **Returning Officer** (2022); Balliol **Football Captain** (2021-22); Balliol **Tennis Team Member** (2022); OU's **Scandinavian Society Member** (2022); **Balliol Choir Member** (2019-20); UF's **Culinary Arts Student Union's Logistic Director** (2015-16)

Selected Presentations:

2024 Singapore; **World Congress for Psychiatric Genetics**, "Polygenic Scores across Environmental Contexts: How Intersectional Factors Affects Predictive Accuracy for Psychiatric Conditions" (Poster)
- **Early Career Investigator Program Awardee; Best Poster Presentation Winner**

2022 Les Houches, France; **Balliol College Reading Course**, "Understanding Artificial Intelligence from the Lens of Wittgenstein" (Oral)

2022 Oxford, UK; **University of Oxford** Medical Image Analysis Seminar Series, "Generative Adversarial Networks for Microscopy Images" (Oral)

2021 Online; **National Institute of Health** Sensory Neuroscience Seminar Series, "Using Machine Learning to Study Starburst Amacrine Cells" (Oral)

2019 Berlin, Germany; **Max Planck / HHMI Connectomics Meeting**, "Automatic Segmentation of Overlapping Neurons Using Convolutional Neural Networks" (Oral)

2018 Orlando, Florida; **17th IEEE Intl. Conference for Machine Learning and Applications**, "Prediction of Sorghum bicolor Genotype from In-situ Images Using Autoencoder-identified SNPs" (Oral)

2017 Anchorage, Alaska; **30th Intl. Joint Conference on Neural Networks**, "Fusing Attention with Visual Question Answering." (Poster)

Selected Publications:

[1] **M. Cudic***, J. Del La Hoz*, et. al "The Impact of Extended COVID-19 Containment Policies on Psychological Distress in the United States: A Causal Investigation", *in Preparation*.

[2] **M. Cudic**, J. Tubbs, T. Ge, J. Smoller "Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk", *submitted to American Journal for Human Genetics*.

[3] **M. Cudic**, J. Diamond, A. Noble "Neuron Segmentation from Low Signal-to-Noise 3D Fluorescent Microscopy Images using Local and Global Scale Fusion," *Fluorescence Imaging of the Brain*, 59.

[4] **M. Cudic**, J. Diamond, A. Noble "Unpaired Mesh-to-Image Translation for 3D Fluorescent Images of Neurons," *Medical Image Analysis,* (2023), 102768.

[5] **M. Cudic**, J. Principe, "Using a Recurrent Kernel Learning Machine for Small-Sample Image Classification," *Jt. Conf. Neural. Netw. (IJCNN)*, Anchorage, AK, (2019) pp. 1-6.

[6] **M. Cudic**, H. Baweja, T. Parhar, S. Nuske, "Prediction of *Sorghum bicolor* Genotype from In-situ Images Using Autoencoder-identified SNPs", *17th IEEE Intl. Conf. Machine Learn. Appl. (ICMLA)*, Orlando, FL, (2018), pp. 23-31.

[7] **M. Cudic**, R. Burt, E. Santana, and J. Principe, "A Flexible Testing Environment for Visual Question Answering with Performance Evaluation," *NeuroComputing,* (2018), 291, pp. 128-135.

[8] R. Burt, **M. Cudic** and J. Principe, "Fusing Attention with Visual Question Answering," *Jt. Conf. Neural. Netw. (IJCNN)*, Anchorage, AK, (2017) pp. 949-953.